Problem Set 4A

Introduction to vectors

Do these after the slides for Section 4A. Textbook Exercise 4A: Questions 12, 13, 15, 16. Answers are on the last page. Show full working in your exercise book.

Skills

1 $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$. Write each vector in terms of $\mathbf{a}$ and $\mathbf{c}$.

(a) $\overrightarrow{OB}$

(b) $\overrightarrow{AC}$

(c) $\overrightarrow{BO}$

2 In triangle OAB, $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. M is the midpoint of OA and N is the midpoint of AB. Find:

(a) $\overrightarrow{OM}$

(b) $\overrightarrow{ON}$

(c) $\overrightarrow{MN}$

3 Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$. P lies on AB with $AP : PB = 2 : 3$. Find $\overrightarrow{OP}$.

4 Simplify:

(a) $3(\mathbf{a} - 2\mathbf{b}) - 2(\mathbf{a} + \mathbf{b})$

(b) $\overrightarrow{AB} + \overrightarrow{BC} - \overrightarrow{AC}$

(c) $\frac{1}{2}(4\mathbf{a} - 6\mathbf{b}) + 3\mathbf{b}$

5 $\mathbf{a}$ and $\mathbf{b}$ are non-zero and not parallel. Find x and y.

(a) $(x - 1)\mathbf{a} + (2y + 1)\mathbf{b} = 3\mathbf{a} + 7\mathbf{b}$

(b) $(x + y)\mathbf{a} + (x - 2y)\mathbf{b} = 7\mathbf{a} + \mathbf{b}$

Exam style

6 $\mathbf{a}$ and $\mathbf{b}$ are not parallel. Let $\mathbf{u} = 2\mathbf{a} + \mathbf{b}$, $\mathbf{v} = \mathbf{a} - 3\mathbf{b}$ and $\mathbf{w} = 7\mathbf{a} - 7\mathbf{b}$. Show that $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent.

7 In triangle OAB, P lies on OA with $OP : PA = 1 : 2$ and Q lies on OB with $OQ : QB = 1 : 2$. Show that PQ is parallel to AB and find $PQ : AB$.

8 $OABC$ is a parallelogram with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OC} = \mathbf{c}$, and M is the midpoint of AB. The line OM meets the diagonal AC at X. Given $\overrightarrow{OX} = \lambda\overrightarrow{OM}$ and $\overrightarrow{AX} = \mu\overrightarrow{AC}$, find λ and μ.

Problem Set 4A: answers

1 (a) $\mathbf{a}+\mathbf{c}$ (b) $\mathbf{c}-\mathbf{a}$ (c) $-\mathbf{a}-\mathbf{c}$

2 (a) $\frac{1}{2}\mathbf{a}$ (b) $\frac{1}{2}(\mathbf{a}+\mathbf{b})$ (c) $\frac{1}{2}\mathbf{b}$

3 $\frac{1}{5}(3\mathbf{a}+2\mathbf{b})$

4 (a) $\mathbf{a}-8\mathbf{b}$ (b) $\mathbf{0}$ (c) $2\mathbf{a}$

5 (a) $x=4,\ y=3$ (b) $x=5,\ y=2$

6 $\mathbf{w}=2\mathbf{u}+3\mathbf{v}$

7 $\overrightarrow{PQ}=\frac{1}{3}\mathbf{b}-\frac{1}{3}\mathbf{a}=\frac{1}{3}\overrightarrow{AB}$, so $PQ \parallel AB$ and $PQ:AB=1:3$

8 $\lambda(\mathbf{a}+\frac{1}{2}\mathbf{c})=(1-\mu)\mathbf{a}+\mu\mathbf{c}$, so $\lambda=\frac{2}{3},\ \mu=\frac{1}{3}$